

The Laguerre-Gaussian Soliton: Optical Vortices as Topological Defects in the Associative Vacuum

Aaron M. Schutz

January 11, 2026

Abstract

In 2010, I investigated the transmission of quantum information using the orbital angular momentum (OAM) of light, successfully generating Laguerre-Gaussian (LG) modes via computer-generated holography [2]. Re-evaluating this work through the lens of *Axiomatic Physical Homeostasis* (APH), I identify the LG mode not merely as a wavefront solution to the paraxial Helmholtz equation, but as a *Topological Soliton* within the electromagnetic field. The characteristic “phase singularity” observed at the beam axis represents a 1D filament of **Stiffness Collapse** ($\beta \rightarrow 0$), where the associative geometry of the vacuum fails. Consequently, the “doughnut” intensity profile is reinterpreted as the vacuum expelling energy from the non-associative core via Geometric Buoyancy. I conclude that encoding information in OAM is equivalent to encoding data in the winding number of the vacuum metric, providing the experimental precursor to the Sedenion-based information topology currently under development.

1 Introduction

Modern quantum information theory relies heavily on the manipulation of photon states. In our 2010 study, “Transmission of Quantum Information via Laguerre Gaussian Modes,” I explored the use of Orbital Angular Momentum (OAM) as a high-density information carrier [2]. I experimentally verified that photons could carry quantized OAM values of $l\hbar$, theoretically allowing for infinite information density per particle.

However, the physical mechanism enforcing the stability of these “twisted” light beams was treated purely as a wave-interference phenomenon. In this retrospective analysis, I apply the APH framework to demonstrate that the *Optical Vortex* is a microscopic analogue to the macroscopic *Event Horizon* described in our recent work on Kerr Black Holes [3]. Both are manifestations of “Stiffness Collapse” in the vacuum geometry.

2 The Paraxial Limit as the Associative Regime

In 2010, I noted that for a particle of spin $s = 1$, a pure state with well-defined spin and OAM quantum numbers is generally impossible due to spin-orbit interaction [2]. To bypass this, I invoked the *Paraxial Approximation*:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} - 2ik\frac{\partial}{\partial z}\right)u(r) = 0 \quad (1)$$

Within APH, the Paraxial Approximation corresponds to the **Weak Buffer Regime** ($\beta \approx 1$). When light propagates with minimal transverse momentum ($k_\perp \ll k_z$), the local geometry remains flat and associative. The “spin-orbit coupling” that arises outside this limit is identified as the **Interaction Hazard** ($\mathcal{A}_{int}$). When the Poynting vector shears significantly against the vacuum lattice, the associative structure breaks down, mixing the internal (Spin) and external (OAM) algebras.

3 The Singularity: A 1D Event Horizon

The defining feature of the LG mode is the scalar field term:

$$u_{np}(r, \theta, z) \propto e^{in\theta} L_p^{|n|}\left(\frac{2r^2}{w^2}\right) \quad (2)$$

In our original analysis, I observed that this term creates a “line of singular phase down the center of the beam” [2].

3.1 Stiffness Collapse ($\beta \rightarrow 0$)

In standard optics, the intensity at $r = 0$ must be zero because the phase is undefined. In APH, this is a topological necessity derived from the stiffness parameter β . At the singularity, the winding number n creates a geometric torsion that the associative vacuum cannot resolve. The local stiffness collapses:

$$\lim_{r \rightarrow 0} \beta(r) = 0 \quad (3)$$

This is effectively a microscopic, 1D Event Horizon. The core of the laser beam contains a filament of non-associative geometry (Sedenion Chaos).

3.2 Geometric Buoyancy

I observed that the intensity profile forms a “doughnut” shape with a dark center [2]. Standard physics attributes this to destructive interference. APH attributes this to **Geometric Buoyancy**. The surrounding vacuum ($\beta \approx 1$) exerts a pressure that expels the electromagnetic flux *out* of the non-associative defect. The photon field effectively “orbits” the defect because it cannot penetrate the region of zero stiffness.

4 Holography as a Boundary Operator

To generate these modes, I employed Computer Generated Holograms (CGH) featuring a characteristic “forked” dislocation pattern [2].

In APH, the diffraction grating functions as a **Boundary Condition Operator**. By forcing the coherent light through the topological dislocation, I impose a constraint on the vacuum geometry. The laser light (Information Flux) is forced to wrap around the defect, acquiring a **Geometric Charge** n . This experimentally validates the APH postulate that topology can constrain energy flow into stable solitons.

5 Conclusion: The Trilogy of Collapse

This reinterpretation unifies our research trajectory into a single coherent narrative regarding the failure modes of vacuum geometry:

1. **Micro-Scale (2010):** The Optical Vortex is a 1D stiffness collapse stabilized by the paraxial limit.
2. **Meso-Scale (2025):** The Chaotic Double Pendulum is a temporary stiffness collapse stabilized by Super-Linear Control [1].
3. **Macro-Scale (2025):** The Kerr Black Hole is a 3D stiffness collapse stabilized by the Sedenion domain wall [3].

I conclude that the “Quantum Information” I sought to transmit in 2010 was, in fact, encoded in the topology of the vacuum metric itself.

References

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